

Digital Retail Banking Platform (DRBP)

Business Requirement Document (BRD)

Document Control

Field	Details
Document Version	1.0
Document Status	Draft
Created By	Shreemudra Satapathy
Created Date	02/03/2025
Last Updated By	Shreemudra Satapathy
Review Date	02/05/2025
Last Updated Date	03/03/2025
Next Review Date	08/06/2025

Table of Content

Digital Retail Banking Platform (DRBP)	1
Table of Content	3
1. Project Overview.....	3
2. Business Problem.....	4
3. Business Objectives	4
4. Scope	5
5. Assumptions & Constraints	6
6. Risk	7
7. Stakeholder Analysis.....	7
8. Requirement Gathering Approach	9
9. Functional Requirements	11
10. Non-Functional Requirements	12
11. Business Rules	12
Module 1: Customer Registration & eKYC	13
Module 2: Login with MFA	13
Module 3: Dashboard.....	13
Module 4: Account Summary.....	13
Module 5: Fund Transfer	13
Module 6: Bill Payment.....	14
Module 7: Fixed Deposit	14
Module 8: Credit Card Management	14
Module 9: Personal Loan Application	14
Module 10: Notifications	15
Module 11: Customer Support	15
Module 12: Admin Portal	15

12. User Stories.....	15
Module 1: Customer Registration & eKYC	15
Module 2: Login with MFA	17
Module 3: Dashboard.....	18
Module 4: Account Summary.....	18
Module 5: Fund Transfer.....	19
Module 6: Bill Payment.....	20
Module 7: Fixed Deposit	21
Module 8: Credit Card Management	22
Module 9: Personal Loan Application	23
Module 10: Notifications	25
Module 11: Customer Support	25
Module 12: Admin Portal	26
13. Acceptance Criteria	29
Module 1: Customer Registration & eKYC	29
Module 2: Login with MFA	30
Module 3: Dashboard.....	30
Module 4: Account Summary.....	30
Module 5: Fund Transfer.....	30
Module 6: Bill Payment.....	31
Module 7: Fixed Deposit	31
Module 8: Credit Card Management	32
Module 9: Personal Loan Application	32
Module 10: Notifications	33
Module 11: Customer Support	33
Module 12: Admin Portal	33
14. Process Flow	34
Process Name	34
Objective	34

Actors	35
Step-by-Step Current Process (20 Steps)	35
Decision Points	36
Manual Activities	36
System Interactions	36
Pain Points and Bottlenecks	36
Business Rules	37
Inputs and Outputs	37
Opportunities for Improvement.....	37
BPMN-Style Process Flow Description.....	38
Pain Points Identified by the Business Analyst.....	39
Recommendations for the To-Be Process	39
Process Name	41
Business Objective	41
Actors	41
Step-by-Step Future Process (18 Steps).....	42
Decision Points	42
Automated System Activities	43
System Integrations	43
Business Rules	43
Inputs and Outputs	43
Exception Scenarios	44
BPMN-Style Process Flow Description.....	44
15. UAT Scenarios	47
Module 1: Customer Registration & eKYC	47
Module 2: Login with MFA	48
Module 3: Dashboard.....	48
Module 4: Account Summary.....	49
Module 5: Fund Transfer.....	49

Module 6: Bill Payment.....	50
Module 7: Fixed Deposit	50
Module 8: Credit Card Management.....	51
Module 9: Personal Loan Application	52
Module 10: Notifications	52
Module 11: Customer Support	53
Module 12: Admin Portal	53
16. Challenge & Solutions	54
17. BA tools & Techniques.....	56

1. Project Overview

Project Name: Digital Retail Banking Platform (DRBP)

Industry: BFSI (Banking, Financial Services, and Insurance)

Client: ABC Bank Ltd.

Project Duration: 10 months

Methodology: Agile (Scrum)

- Sprint Duration: 2 Weeks
- Total Sprints: 20

Role: Senior Business Analyst

Responsibilities:

- Conduct stakeholder interviews
- Gather business requirements
- Facilitate requirement workshops
- Document BRD and FRD
- Write user stories and acceptance criteria
- Create AS-IS and TO-BE process flows
- Prepare wireframe requirements
- Manage the Requirement Traceability Matrix (RTM)
- Support Sprint Planning and Sprint Review
- Collaborate with developers, testers, and Product Owner
- Support User Acceptance Testing (UAT)
- Assist with production rollout

Project Summary:

The Digital Retail Banking Platform is designed to provide customers with secure and convenient banking services through web and mobile applications. The platform enables

digital account onboarding, account management, fund transfers, bill payments, deposits, card services, and loan applications. The project aims to enhance customer experience, reduce dependency on branch visits, automate manual processes, and improve operational efficiency while ensuring compliance with banking regulations and security standards.

2. Business Problem

ABC Bank has experienced a significant increase in customers using digital channels, but its existing online banking platform is outdated, slow, and lacks key self-service features. Customers must visit branches for account opening, KYC verification, fixed deposits, and loan applications, resulting in long wait times and high operational costs.

The bank also faces challenges with:

- Poor customer experience due to multiple disconnected systems.
- High customer support calls for routine banking requests.
- Delays in digital onboarding because of manual KYC verification.
- Low adoption of digital services.
- Security concerns related to authentication and fraud detection.

To remain competitive with digital-first banks and fintech companies, the bank plans to implement a modern Digital Retail Banking Platform.

3. Business Objectives

Primary Objective

Develop a secure, user-friendly, and scalable Digital Retail Banking Platform that enables customers to perform everyday banking services online, reducing dependency on branch visits while improving customer satisfaction and operational efficiency.

Detailed Business Objectives

1. Increase digital adoption by enabling customers to access banking services through web and mobile applications.
2. Reduce branch visits by **40%** within **12 months** by offering end-to-end digital self-service capabilities.

3. Decrease customer onboarding time from **2 days to less than 15 minutes** using digital eKYC and online account opening.
4. Enable secure and real-time transactions, including fund transfers, bill payments, deposits, and card management.
5. Improve customer satisfaction by providing an intuitive user experience and 24×7 banking availability.
6. Reduce operational costs by automating manual banking processes and minimizing paper-based workflows.
7. Strengthen security through Multi-Factor Authentication (MFA), encryption, fraud detection, and real-time transaction monitoring.
8. Ensure compliance with banking regulations, KYC, AML, and data privacy requirements.
9. Provide personalized banking services such as transaction insights, spending summaries, and tailored product recommendations.
10. Improve employee productivity by reducing manual processing and allowing branch staff to focus on higher-value customer interactions.

4. Scope

The project aims to design and implement a Digital Retail Banking Platform that enables retail customers to securely access banking services through web and mobile applications. The platform will provide digital self-service capabilities for account management, payments, transfers, deposits, loans, and customer support while ensuring security, regulatory compliance, and a seamless user experience.

In Scope

- Digital account opening via eKYC (document upload + video KYC)
- Secure login with multi-factor authentication (OTP/biometric)
- Consolidated post-login dashboard with account, card, and alert summary
- Account summary — balances, statements, transaction history
- Fund transfer (NEFT, RTGS, IMPS) with beneficiary management
- Bill payment with biller registration and payment history
- Fixed deposit booking, tenure/interest calculation, premature closure
- Credit card activation, limit view, statement access, block/replace requests
- Personal loan eligibility check, application, and status tracking
- Transactional and service notifications (SMS/email/in-app)

- Customer support via ticketing and chatbot/FAQ assistance
- Admin portal for internal user, application, and platform management

Out of Scope

- Wealth management and investment product services (e.g., mutual funds, insurance cross-sell) .
- Corporate and business banking services (platform is retail-customer only) .
- Replacement of existing IVR and call-centre systems (platform will integrate with, not replace, these during transition)
- Self-service credit limit increases above the bank's pre-approved threshold (requires separate approval workflow per BR-25)
- Merchant/POS-side payment processing
- Third-party account aggregation (e.g., viewing accounts held at other banks)

5. Assumptions & Constraints

Assumptions

- Customers have a smartphone or computer with internet access and a valid government ID for eKYC.
- Core banking system (CBS) exposes APIs for real-time balance, transaction, and account data.
- NPCI/UPI and NEFT/RTGS gateways are accessible through existing bank infrastructure.
- Relationship managers and support staff will be trained before platform rollout.
- Existing customer data can be migrated from legacy systems without loss.

Constraints

- Must comply with RBI's Master Direction on Digital Payment Security Controls and KYC norms.
- MFA must add no more than one extra step to the login flow, per the bank's UX guidelines.
- Legacy data migration is dependent on the core banking vendor's export timelines.
- Platform must integrate with existing IVR and call-centre systems during transition, not replace them outright.

- Delivery timeline is fixed by the bank's board-approved digital transformation roadmap.

6. Risk

Risk	Impact	Likelihood	Mitigation
CBS API integration delays	Delays core banking data availability, pushing back launch	Medium	Early integration testing with core banking vendor; phased API rollout
Customer resistance to eKYC/digital onboarding	Slower adoption than projected, undermining onboarding targets	Medium	In-app guidance, branch-assisted digital onboarding option during transition
MFA-related login friction	Increased drop-off or support calls if OTP delivery is unreliable	Medium	Multiple MFA channels (SMS + biometric); fallback OTP resend flow
Regulatory non-compliance (RBI KYC/security norms)	Legal penalties, mandated platform changes post-launch	Low	Compliance officer sign-off at each phase; periodic regulatory review
Data migration errors from legacy systems	Incorrect balances/history shown to customers, trust erosion	Medium	Data validation and reconciliation before cutover; parallel run period
Fraud attempts during transition period	Financial loss, reputational damage	Medium	Real-time transaction monitoring, fraud team review of flagged logins
Scope creep from stakeholder requests	Timeline and budget overrun	High	Change request process tied to board-approved roadmap; BA-led scope sign-off
Third-party payment gateway downtime	Failed transactions, customer complaints	Low	SLA-backed gateway contracts; fallback to secondary provider

7. Stakeholder Analysis

Stakeholder	Role	Interest/Stake	Influence	Involvement
Retail Customer	End user	Fast onboarding, secure	Low (indirect, via feedback/adoption)	End-to-end platform user

		access, self-service banking		
Relationship Manager	Sales/service	Digital tools to support onboarding, FD, and loan follow-ups	Medium	Consulted during onboarding and lending flow design
Loan Processing Team	Operations	Efficient, accurate digital loan application intake	Medium	Reviews and processes applications via Admin Portal
Customer Support Executive	Support	Reduced call volume; tools to resolve escalated tickets	Medium	Uses ticketing/chatbot module; provides UAT feedback
Fraud & Risk Team	Risk management	Strong authentication, transaction monitoring	High	Defines MFA and fraud-flagging requirements
Compliance Officer	Regulatory oversight	Adherence to RBI KYC and security norms	High	Sign-off on eKYC and security design
IT/Core Banking Team	Technical delivery	Stable CBS integration, system uptime	High	Owns API integration and platform infrastructure
System Administrator	Platform operations	Manageable user roles, monitoring, configuration	Medium	Primary user of Admin Portal
Bank Leadership/Board	Sponsor	ROI on digital transformation roadmap, customer retention	High	Approves scope, budget, and timeline
Business Analyst	Project delivery	Accurate requirements, stakeholder alignment	Medium	Elicits requirements, manages scope,

				documents traceability
--	--	--	--	---------------------------

8. Requirement Gathering Approach

1. Overview

This document outlines the approach used to elicit, analyze, and validate requirements for the Digital Retail Banking Platform. Given the platform spans 12 modules involving multiple internal teams (compliance, fraud & risk, IT, loan processing) and external-facing customer journeys, a mixed-method elicitation approach was adopted — combining stakeholder interviews, workshops, document analysis, and process observation — to ensure requirements were both business-accurate and technically feasible before being finalized into the Functional and Non-Functional Requirements.

2. Objectives of Requirement Gathering

- Identify pain points in the bank's existing digital channels (net banking, mobile app, IVR) directly from the teams and customers affected by them.
- Capture functional needs for each of the 12 modules from the stakeholders who will use or be impacted by them.
- Surface regulatory and security constraints early, before design work begins.
- Validate assumptions made in the Project Overview and Scope documents against real operational input.

3. Stakeholder Engagement

Stakeholder engagement was structured around the stakeholder groups identified in the Stakeholder Analysis, with the elicitation method matched to each group's role and availability:

Stakeholder	Engagement Method	Purpose
Retail Customers	Surveys + usability interviews	Understand pain points in current onboarding, transfers, and support experience
Relationship Managers	Focus group workshop	Capture friction points in FD/loan sales that require branch follow-up

Loan Processing Team	One-on-one interviews	Document current loan intake, document verification, and approval workflow
Customer Support Executives	Shadowing + interviews	Identify recurring, high-volume queries suitable for self-service or chatbot deflection
Fraud & Risk Team	Workshop	Define MFA requirements and fraud-flagging rules
Compliance Officer	Document review + interview	Confirm RBI KYC and security norms that must be built into eKYC and login flows
IT/Core Banking Team	Technical workshop	Assess CBS API availability and integration constraints
Bank Leadership	Kickoff and sign-off meetings	Align on business objectives, scope boundaries, and success criteria

4. Elicitation Techniques Used

- **Stakeholder Interviews** — one-on-one sessions with Loan Processing, Compliance, and IT teams to capture role-specific requirements and constraints.
- **Workshops** — joint sessions with Fraud & Risk and IT teams to define MFA flow and security requirements collaboratively.
- **Document Analysis** — review of existing net banking portal specifications, RBI KYC guidelines, and past customer complaint logs.
- **Process Observation** — shadowing customer support executives handling live calls to identify which queries could be moved to self-service.
- **Surveys** — sent to a sample of existing customers to gauge digital adoption barriers and feature priorities.
- **Prototype Walkthroughs** — low-fidelity wireframes reviewed with Relationship Managers and Support Executives to validate assumptions before finalizing FRs.

5. Requirement Validation

- Draft requirements were reviewed in a **sign-off workshop** with Compliance, Fraud & Risk, and IT stakeholders to confirm feasibility and regulatory alignment.

- Functional requirements were traced back to specific stakeholder inputs to ensure no requirement was added without a documented source.
- Ambiguous or conflicting requirements (e.g., MFA friction vs. security strictness) were resolved through a follow-up workshop before being finalized in the FR/NFR document.

6. Tools Used

- Requirement documentation: structured user story templates (as used in the User Stories document)
- Traceability: requirements-to-stakeholder mapping maintained in a traceability matrix
- Prototyping: wireframes for key journeys (onboarding, fund transfer, loan application)

9. Functional Requirements

ID	Module	Requirement
FR-01	Customer Registration & eKYC	System shall allow customers to register using PAN/Aadhaar-based eKYC with document upload and video verification
FR-02	Customer Registration & eKYC	System shall validate uploaded documents and flag mismatches for manual review
FR-03	Login with MFA	System shall authenticate users via credentials followed by OTP or biometric verification
FR-04	Login with MFA	System shall lock the account after 3 consecutive failed login attempts
FR-05	Dashboard	System shall display a consolidated view of account balances, recent transactions, and active alerts on login
FR-06	Account Summary	System shall allow customers to view balances, download statements, and search transaction history by date/type
FR-07	Fund Transfer	System shall support NEFT, RTGS, and IMPS transfers with beneficiary addition and daily limit checks
FR-08	Fund Transfer	System shall allow customers to schedule one-time or recurring transfers
FR-09	Bill Payment	System shall allow biller registration and payment with confirmation and receipt generation
FR-10	Fixed Deposit	System shall allow FD booking with tenure and interest calculation, and support premature closure requests

FR-11	Credit Card Management	System shall allow card activation, limit viewing, statement access, and block/replacement requests
FR-12	Personal Loan Application	System shall allow customers to check loan eligibility, submit applications with documents, and track status
FR-13	Notifications	System shall send transactional and service alerts via SMS, email, and in-app notification
FR-14	Customer Support	System shall allow customers to raise support tickets and access FAQ/chatbot assistance
FR-15	Admin Portal	System shall allow administrators to manage customer records, review applications, and configure platform settings
FR-16	Admin Portal	System shall provide role-based access control for internal users

10. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Dashboard and Account Summary pages shall load within 3 seconds under normal load
NFR-02	Security	All customer data in transit and at rest shall be encrypted (TLS 1.2+, AES-256)
NFR-03	Security	MFA shall be enforced on all customer logins without exception
NFR-04	Availability	Platform shall maintain 99.9% uptime, excluding scheduled maintenance windows
NFR-05	Compliance	System shall comply with RBI's Master Direction on Digital Payment Security Controls and KYC norms
NFR-06	Scalability	System shall support concurrent access by up to 500,000 active users without degradation
NFR-07	Usability	Core transactions (login, transfer, bill payment) shall be completable within 4 screens or fewer
NFR-08	Auditability	All financial transactions and admin actions shall be logged with timestamp and user ID for audit purposes
NFR-09	Compatibility	Platform shall be accessible via major web browsers and on Android/iOS mobile devices
NFR-10	Recoverability	System shall support data backup and recovery with a maximum tolerable downtime of 4 hours
NFR-11	Usability/Security	MFA shall add no more than one additional step to the standard login flow, regardless of the second-factor method chosen (OTP or biometric)

11. Business Rules

Module 1: Customer Registration & eKYC

- BR-01: Customer must provide a valid PAN and Aadhaar number; the system shall reject registration if either fails UIDAI/NSDL validation.
- BR-02: Customer must be at least 18 years old to open an individual savings account.
- BR-03: Uploaded documents must be JPEG, PNG, or PDF, and must not exceed 5MB per file.
- BR-04: A mismatch between the uploaded document and the ID database shall route the application to manual compliance review — it shall not be auto-approved.
- BR-05: One Aadhaar/PAN combination shall be linked to only one active retail account.

Module 2: Login with MFA

- BR-06: Every login shall require a second authentication factor (OTP or biometric); single-factor login is not permitted.
- BR-07: An OTP shall expire 5 minutes after issuance and cannot be reused.
- BR-08: The account shall lock after 3 consecutive failed MFA attempts and require identity re-verification to unlock.
- BR-09: A new device or location login shall trigger an additional verification step.

Module 3: Dashboard

- BR-10: The dashboard shall display only accounts and cards linked to the logged-in customer's profile.
- BR-11: Balance figures shown on the dashboard must reflect real-time data from the core banking system, not cached values older than 5 minutes.

Module 4: Account Summary

- BR-12: Statement generation shall be limited to a maximum range of 3 years from the current date.
- BR-13: Statements shall be available for download only after successful MFA-verified login.

Module 5: Fund Transfer

- BR-14: Daily transfer limits shall apply per customer, per transfer mode (NEFT/RTGS/IMPS), as configured by the bank.
- BR-15: A transfer request exceeding the available balance shall be rejected before submission to the payment gateway.
- BR-16: A new beneficiary shall be subject to a cooling period (e.g., 30 minutes to 24 hours) before the first transfer can be made to them.
- BR-17: RTGS transfers shall only be permitted for amounts of ₹2,00,000 or above; amounts below this threshold shall default to NEFT or IMPS.

Module 6: Bill Payment

- BR-18: A biller must be registered with the Bharat Bill Payment System (BBPS) to be eligible for listing.
- BR-19: A bill payment shall not be processed twice for the same biller reference number within 24 hours, to prevent duplicate payment.

Module 7: Fixed Deposit

- BR-20: Minimum FD booking amount shall be ₹1,000; minimum tenure shall be 7 days.
- BR-21: Premature closure shall attract a penalty as per the bank's prevailing FD closure policy, deducted from the payout amount.
- BR-22: Interest earned on FDs shall be subject to TDS as per applicable income tax rules once the threshold is crossed.

Module 8: Credit Card Management

- BR-23: A credit card shall only be activated after successful MFA verification tied to the registered mobile number.
- BR-24: A blocked card shall not be reactivated; only a replacement card issuance is permitted.
- BR-25: Credit limit changes shall require a separate approval workflow and cannot be self-serviced above the bank's pre-approved threshold.

Module 9: Personal Loan Application

- BR-26: Loan eligibility shall be computed based on declared income, existing obligations, and credit bureau (CIBIL) score.
- BR-27: An applicant with an active loan default on record shall be auto-rejected at the eligibility check stage.
- BR-28: Loan disbursement shall only occur after all required documents are verified and approved by the processing team.

Module 10: Notifications

- BR-29: A transactional alert shall be sent within 1 minute of the triggering event, regardless of channel.
- BR-30: A customer shall be able to opt out of promotional notifications but not transactional/security alerts.

Module 11: Customer Support

- BR-31: A support ticket unresolved after 48 hours shall be automatically escalated to a senior support agent.
- BR-32: Chatbot interactions unresolved after 2 failed attempts shall be automatically routed to a live agent.

Module 12: Admin Portal

- BR-33: Internal user access shall be governed by role-based permissions; no internal user shall have access beyond their assigned role.
- BR-34: All administrative actions (user creation, role changes, application approvals) shall be logged with timestamp and user ID for audit purposes.
- BR-35: Audit logs shall be retained for a minimum of 7 years, in line with RBI record-retention norms.

12. User Stories

Module 1: Customer Registration & eKYC

US-01

- **Epic:** Digital Onboarding
- **Module:** Customer Registration & eKYC
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to register for an account using Aadhaar/PAN-based eKYC, so that I can open an account without visiting a branch.
- **Business Value:** Reduces onboarding time from 2-3 days to same-day, improving acquisition and reducing branch load.
- **Priority:** High
- **Story Points:** 8
- **Acceptance Criteria:**
 - Given a customer starts registration, when they enter their Aadhaar/PAN number, then the system validates it against the UIDAI/NSDL database in real time.
 - Given valid identity details, when the customer uploads required documents, then the system accepts JPEG/PNG/PDF files up to 5MB each.
 - Given all documents are submitted, when verification completes successfully, then the customer receives account activation confirmation within the same business day.
- **Dependencies:** Integration with UIDAI/NSDL verification APIs.
- **Assumptions:** Customer has a valid, linked mobile number for Aadhaar OTP verification.

US-02

- **Epic:** Digital Onboarding
- **Module:** Customer Registration & eKYC
- **User Role:** Compliance Officer
- **User Story:** As a compliance officer, I want mismatched KYC documents to be automatically flagged and routed to a manual review queue, so that I can verify borderline applications before they're approved.
- **Business Value:** Preserves regulatory compliance without slowing down the majority of standard applications.
- **Priority:** High
- **Story Points:** 5
- **Acceptance Criteria:**

- Given a document is uploaded, when the system detects a name/DOB mismatch with the ID database, then the application is routed to a manual review queue.
- Given an application is in the review queue, when the compliance officer approves or rejects it, then the customer is notified within 24 hours.
- **Dependencies:** Document verification service, Admin Portal review queue.
- **Assumptions:** Compliance team reviews the queue within the 24-hour notification window.

Module 2: Login with MFA

US-03

- **Epic:** Secure Access
- **Module:** Login with MFA
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to log in using my password followed by an OTP or biometric check, so that my account stays secure from unauthorized access.
- **Business Value:** Reduces account takeover fraud and meets RBI security mandates.
- **Priority:** High
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given correct credentials are entered, when the system prompts for a second factor, then the customer can choose OTP (SMS) or biometric authentication.
 - Given an incorrect OTP is entered 3 times, when the limit is exceeded, then the account is temporarily locked and the customer is notified.
- **Dependencies:** SMS gateway and biometric SDK integration.
- **Assumptions:** Customer's registered mobile number is active and reachable.

US-04

- **Epic:** Secure Access
- **Module:** Login with MFA
- **User Role:** Fraud & Risk Team

- **User Story:** As a fraud & risk analyst, I want failed login attempts and MFA bypass attempts logged and flagged, so that I can investigate potential account compromise.
- **Business Value:** Enables early fraud detection and reduces financial loss exposure.
- **Priority:** Medium
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given repeated failed MFA attempts occur on an account, when the threshold is crossed, then an alert is generated for the fraud team.
 - Given an alert is raised, when the analyst reviews it, then they can temporarily suspend the account pending investigation.
- **Dependencies:** Admin Portal access to fraud alert dashboard.
- **Assumptions:** Fraud team has 24/7 monitoring coverage.

Module 3: Dashboard

US-05

- **Epic:** Account Overview
- **Module:** Dashboard
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to see a consolidated view of my accounts, cards, and recent activity right after login, so that I don't have to navigate multiple screens to check my financial status.
- **Business Value:** Improves customer experience and reduces support queries for basic account status checks.
- **Priority:** High
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given a customer logs in successfully, when the dashboard loads, then it displays account balances, linked cards, and the 5 most recent transactions.
 - Given a customer has active alerts (e.g., low balance, FD maturity), when the dashboard loads, then these alerts appear prominently at the top.
- **Dependencies:** CBS API for real-time balance data (FR-05).
- **Assumptions:** Customer has at least one active account or card.

Module 4: Account Summary

US-06

- **Epic:** Account Servicing
- **Module:** Account Summary
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to view and download my account statements for a selected date range, so that I can track my finances or use them for record-keeping.
- **Business Value:** Reduces branch/call-centre requests for statement copies.
- **Priority:** High
- **Story Points:** 3
- **Acceptance Criteria:**
 - Given a customer selects a date range, when they request a statement, then the system generates it in PDF format within 10 seconds.
 - Given a statement is generated, when the customer chooses to download it, then it is available for at least 24 hours in their session.
- **Dependencies:** CBS transaction history API.
- **Assumptions:** Transaction history is available for at least the past 3 years.

Module 5: Fund Transfer

US-07

- **Epic:** Payments & Transfers
- **Module:** Fund Transfer
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to transfer funds via NEFT, RTGS, or IMPS to a registered beneficiary, so that I can make payments without visiting a branch.
- **Business Value:** Reduces branch transaction volume and increases digital engagement.
- **Priority:** High
- **Story Points:** 8
- **Acceptance Criteria:**

- Given a beneficiary is added and verified, when the customer initiates a transfer within daily limits, then the transfer is processed and a confirmation is displayed.
 - Given a transfer exceeds the daily limit, when submitted, then the system rejects it with an explanatory message.
- **Dependencies:** NPCI/UPI and NEFT/RTGS gateway integration.
- **Assumptions:** Beneficiary bank supports the selected transfer mode.

US-08

- **Epic:** Payments & Transfers
- **Module:** Fund Transfer
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to schedule recurring transfers to a beneficiary, so that I don't have to manually repeat regular payments like rent or EMIs.
- **Business Value:** Increases platform stickiness and reduces missed payment complaints.
- **Priority:** Medium
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given a customer sets up a recurring transfer, when the scheduled date arrives, then the transfer executes automatically if sufficient balance exists.
 - Given insufficient balance on the scheduled date, when the transfer fails, then the customer is notified immediately.
- **Dependencies:** US-07 (fund transfer core functionality).
- **Assumptions:** Customer maintains sufficient balance for scheduled transfers.

Module 6: Bill Payment

US-09

- **Epic:** Payments & Transfers
- **Module:** Bill Payment
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to register billers and pay utility bills through the platform, so that I can manage all my payments in one place.

- **Business Value:** Increases transaction volume on the platform and reduces reliance on third-party bill payment apps.
- **Priority:** Medium
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given a customer searches for a biller, when they register it, then it appears in their saved billers list.
 - Given a bill payment is made, when the transaction completes, then a receipt is generated and stored in payment history.
- **Dependencies:** Biller aggregator/BBPS integration.
- **Assumptions:** Biller is registered with the Bharat Bill Payment System (BBPS).

Module 7: Fixed Deposit

US-10

- **Epic:** Deposit Products
- **Module:** Fixed Deposit
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to book a fixed deposit online by choosing the amount and tenure, so that I can invest without visiting a branch.
- **Business Value:** Increases digital adoption of deposit products and reduces branch dependency.
- **Priority:** High
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given a customer enters an amount and tenure, when they confirm booking, then the system displays the applicable interest rate and maturity value before final confirmation.
 - Given an FD is booked, when the transaction completes, then funds are debited from the linked account and an e-certificate is generated.
- **Dependencies:** CBS FD product rules engine.
- **Assumptions:** Customer has sufficient balance in the linked account.

US-11

- **Epic:** Deposit Products
- **Module:** Fixed Deposit

- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to request premature closure of my fixed deposit, so that I can access funds early if needed.
- **Business Value:** Improves customer flexibility and satisfaction.
- **Priority:** Medium
- **Story Points:** 3
- **Acceptance Criteria:**
 - Given a customer requests premature closure, when the request is submitted, then the system displays the applicable penalty and revised payout amount.
 - Given the customer confirms closure, when processed, then funds are credited to the linked account within 1 business day.
- **Dependencies:** US-10 (FD booking).
- **Assumptions:** Premature closure penalty rates are pre-configured in the system.

Module 8: Credit Card Management

US-12

- **Epic:** Card Servicing
- **Module:** Credit Card Management
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to activate my new credit card and view my credit limit, so that I can start using it without calling customer support.
- **Business Value:** Reduces card activation-related call-centre volume.
- **Priority:** High
- **Story Points:** 3
- **Acceptance Criteria:**
 - Given a new card is issued, when the customer enters the card and CVV details, then the system activates it after MFA verification.
 - Given the card is active, when the customer views card details, then the current limit and available credit are displayed.
- **Dependencies:** Card management system integration.
- **Assumptions:** Card has been dispatched and received by the customer.

US-13

- **Epic:** Card Servicing

- **Module:** Credit Card Management
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to block my credit card and request a replacement if it's lost or stolen, so that I can prevent unauthorized use immediately.
- **Business Value:** Reduces fraud losses from lost/stolen cards.
- **Priority:** High
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given a customer selects "block card," when confirmed via MFA, then the card is blocked instantly and cannot be used for transactions.
 - Given a card is blocked, when the customer requests a replacement, then a new card is dispatched within 5-7 business days.
- **Dependencies:** Card management system, courier/dispatch integration.
- **Assumptions:** Customer's registered address is current and accurate.

Module 9: Personal Loan Application

US-14

- **Epic:** Lending
- **Module:** Personal Loan Application
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to check my personal loan eligibility online, so that I know my chances before submitting a full application.
- **Business Value:** Reduces application abandonment by setting expectations early.
- **Priority:** High
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given a customer enters income and existing obligation details, when they request an eligibility check, then the system displays an indicative loan amount and interest rate.
 - Given eligibility is displayed, when the customer proceeds, then they are directed to the full application form.
- **Dependencies:** Credit bureau (CIBIL) integration.
- **Assumptions:** Customer provides accurate income information.

US-15

- **Epic:** Lending
- **Module:** Personal Loan Application
- **User Role:** Loan Processing Team
- **User Story:** As a loan processing officer, I want to review submitted loan applications and documents in a queue, so that I can approve or reject them efficiently.
- **Business Value:** Speeds up loan turnaround time and reduces manual coordination.
- **Priority:** High
- **Story Points:** 8
- **Acceptance Criteria:**
 - Given an application is submitted, when it appears in the processing queue, then the officer can view all documents and eligibility data in one screen.
 - Given the officer approves or rejects an application, when the decision is saved, then the customer is notified and the status updates in their application tracker.
- **Dependencies:** Admin Portal integration, document storage system.
- **Assumptions:** Officer has access to CIBIL score data for decisioning.

US-16

- **Epic:** Lending
- **Module:** Personal Loan Application
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to track the status of my loan application in real time, so that I know what stage it's at without calling support.
- **Business Value:** Reduces status-check calls to the support centre.
- **Priority:** Medium
- **Story Points:** 3
- **Acceptance Criteria:**
 - Given an application is submitted, when the customer views the tracker, then it shows the current stage (Submitted, Under Review, Approved, Disbursed, or Rejected).
 - Given the status changes, when an update occurs, then the customer receives a notification.
- **Dependencies:** US-15, Notifications module.

- **Assumptions:** Status updates are made promptly by the processing team.

Module 10: Notifications

US-17

- **Epic:** Customer Communication
- **Module:** Notifications
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to receive alerts for transactions, due dates, and account activity via SMS, email, or in-app notification, so that I stay informed without actively checking the app.
- **Business Value:** Improves transparency and reduces missed-payment/fraud-related complaints.
- **Priority:** High
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given a transaction occurs on the account, when it is processed, then a notification is sent within 1 minute via the customer's preferred channel.
 - Given a customer has upcoming FD maturity or bill due dates, when the due date approaches, then a reminder is sent 3 days in advance.
- **Dependencies:** SMS/email gateway integration.
- **Assumptions:** Customer has provided and verified contact details.

Module 11: Customer Support

US-18

- **Epic:** Customer Service
- **Module:** Customer Support
- **User Role:** Retail Customer
- **User Story:** As a retail customer, I want to raise a support ticket and track its status, so that I can get help without calling the branch or waiting on hold.
- **Business Value:** Reduces inbound call volume and gives customers self-service visibility into issue resolution.
- **Priority:** Medium

- **Story Points:** 3
- **Acceptance Criteria:**
 - Given a customer submits a ticket with issue details, when it's created, then a ticket ID is generated and the customer can track its status.
 - Given a ticket remains unresolved for 48 hours, when the SLA breach occurs, then it is auto-escalated to a senior support agent.
- **Dependencies:** Ticketing system integration.
- **Assumptions:** Customer is logged in when raising a ticket.

US-19

- **Epic:** Customer Service
- **Module:** Customer Support
- **User Role:** Customer Support Executive
- **User Story:** As a customer support executive, I want access to a chatbot handling common FAQs, so that I can focus on complex escalated queries instead of repetitive questions.
- **Business Value:** Reduces executive workload and improves first-response time for simple queries.
- **Priority:** Medium
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given a customer asks a common question (e.g., "how to reset MFA"), when the chatbot identifies the intent, then it provides a resolution without human intervention.
 - Given the chatbot cannot resolve a query, when it detects this, then it offers to escalate the issue to a live agent.
- **Dependencies:** Chatbot/NLP engine integration.
- **Assumptions:** FAQ content is maintained and updated regularly.

Module 12: Admin Portal

US-20

- **Epic:** Platform Administration
- **Module:** Admin Portal
- **User Role:** System Administrator

- **User Story:** As a system administrator, I want to manage internal user roles and access permissions, so that employees only have access to the functions relevant to their role.
- **Business Value:** Reduces risk of unauthorized access to sensitive customer or financial data.
- **Priority:** High
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given an administrator creates a new internal user, when a role is assigned, then access is restricted to permissions mapped to that role.
 - Given a user's role changes, when updated, then access permissions are revised immediately.
- **Dependencies:** NFR-08 (audit logging).
- **Assumptions:** Role definitions are finalized before platform launch.

US-21

- **Epic:** Platform Administration
- **Module:** Admin Portal
- **User Role:** System Administrator
- **User Story:** As a system administrator, I want to view platform health metrics (uptime, transaction volumes, error rates), so that I can proactively identify and resolve issues.
- **Business Value:** Improves platform reliability and reduces downtime impact on customers.
- **Priority:** Medium
- **Story Points:** 8
- **Acceptance Criteria:**
 - Given the admin portal loads, when the administrator views the monitoring dashboard, then it displays real-time uptime, error rate, and transaction volume metrics.
 - Given a system anomaly is detected (e.g., error rate spike), when the threshold is breached, then an alert is sent to the IT team.
- **Dependencies:** Platform monitoring/logging infrastructure.
- **Assumptions:** Monitoring tools are integrated at the infrastructure layer.

US-22

- **Epic:** Platform Administration
- **Module:** Admin Portal
- **User Role:** Compliance Officer
- **User Story:** As a compliance officer, I want to generate audit reports of customer onboarding and transaction activity, so that I can demonstrate regulatory compliance during audits.
- **Business Value:** Reduces audit preparation time and regulatory risk.
- **Priority:** High
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given a compliance officer selects a date range, when they generate a report, then it includes onboarding, transaction, and MFA logs for that period.
 - Given a report is generated, when exported, then it is available in PDF/Excel format.
- **Dependencies:** NFR-08 (audit logging), Admin Portal reporting module.
- **Assumptions:** All relevant data is retained per regulatory retention requirements.

US-23

- **Epic:** Platform Administration
- **Module:** Admin Portal
- **User Role:** Fraud & Risk Team
- **User Story:** As a fraud & risk analyst, I want a dashboard showing flagged accounts and suspicious transaction patterns, so that I can investigate and act quickly.
- **Business Value:** Enables faster fraud detection and reduces financial loss.
- **Priority:** High
- **Story Points:** 8
- **Acceptance Criteria:**
 - Given a transaction matches a fraud rule (e.g., unusual amount/location), when flagged, then it appears on the fraud dashboard with relevant details.
 - Given an analyst reviews a flagged transaction, when they mark it as fraudulent, then the account is suspended pending further review.
- **Dependencies:** US-04 (login fraud flagging), fraud rules engine.
- **Assumptions:** Fraud detection rules are pre-configured and periodically tuned.

US-24

- **Epic:** Platform Administration
- **Module:** Admin Portal
- **User Role:** Loan Processing Team
- **User Story:** As a loan processing officer, I want a consolidated view of all pending loan applications with filters by status and date, so that I can prioritize my workload effectively.
- **Business Value:** Improves loan processing turnaround time.
- **Priority:** Medium
- **Story Points:** 5
- **Acceptance Criteria:**
 - Given the officer opens the loan queue, when filters are applied (status, date, amount), then only matching applications are displayed.
 - Given an application is selected, when opened, then all supporting documents and eligibility data are visible on one screen.
- **Dependencies:** US-15.
- **Assumptions:** Application data is synced in real time from the customer-facing module.

13. Acceptance Criteria

Module 1: Customer Registration & eKYC

US-01 — Register via eKYC

- Given a customer starts registration, when they enter their Aadhaar/PAN number, then the system validates it against the UIDAI/NSDL database in real time.
- Given valid identity details, when the customer uploads required documents, then the system accepts JPEG/PNG/PDF files up to 5MB each.
- Given all documents are submitted, when verification completes successfully, then the customer receives account activation confirmation within the same business day.

US-02 — Flag mismatched KYC documents

- Given a document is uploaded, when the system detects a name/DOB mismatch with the ID database, then the application is routed to a manual review queue.

- Given an application is in the review queue, when the compliance officer approves or rejects it, then the customer is notified within 24 hours.

Module 2: Login with MFA

US-03 — Login with OTP/biometric

- Given correct credentials are entered, when the system prompts for a second factor, then the customer can choose OTP (SMS) or biometric authentication.
- Given an incorrect OTP is entered 3 times, when the limit is exceeded, then the account is temporarily locked and the customer is notified.

US-04 — Flag failed login/MFA attempts

- Given repeated failed MFA attempts occur on an account, when the threshold is crossed, then an alert is generated for the fraud team.
- Given an alert is raised, when the analyst reviews it, then they can temporarily suspend the account pending investigation.

Module 3: Dashboard

US-05 — Consolidated post-login view

- Given a customer logs in successfully, when the dashboard loads, then it displays account balances, linked cards, and the 5 most recent transactions.
- Given a customer has active alerts (e.g., low balance, FD maturity), when the dashboard loads, then these alerts appear prominently at the top.

Module 4: Account Summary

US-06 — View and download statements

- Given a customer selects a date range, when they request a statement, then the system generates it in PDF format within 10 seconds.
- Given a statement is generated, when the customer chooses to download it, then it is available for at least 24 hours in their session.

Module 5: Fund Transfer

US-07 — Transfer via NEFT/RTGS/IMPS

- Given a beneficiary is added and verified, when the customer initiates a transfer within daily limits, then the transfer is processed and a confirmation is displayed.
- Given a transfer exceeds the daily limit, when submitted, then the system rejects it with an explanatory message.

US-08 — Schedule recurring transfers

- Given a customer sets up a recurring transfer, when the scheduled date arrives, then the transfer executes automatically if sufficient balance exists.
- Given insufficient balance on the scheduled date, when the transfer fails, then the customer is notified immediately.

Module 6: Bill Payment

US-09 — Register billers and pay bills

- Given a customer searches for a biller, when they register it, then it appears in their saved billers list.
- Given a bill payment is made, when the transaction completes, then a receipt is generated and stored in payment history.

Module 7: Fixed Deposit

US-10 — Book a fixed deposit

- Given a customer enters an amount and tenure, when they confirm booking, then the system displays the applicable interest rate and maturity value before final confirmation.
- Given an FD is booked, when the transaction completes, then funds are debited from the linked account and an e-certificate is generated.

US-11 — Premature FD closure

- Given a customer requests premature closure, when the request is submitted, then the system displays the applicable penalty and revised payout amount.

- Given the customer confirms closure, when processed, then funds are credited to the linked account within 1 business day.

Module 8: Credit Card Management

US-12 — Activate card and view limit

- Given a new card is issued, when the customer enters the card and CVV details, then the system activates it after MFA verification.
- Given the card is active, when the customer views card details, then the current limit and available credit are displayed.

US-13 — Block and replace card

- Given a customer selects "block card," when confirmed via MFA, then the card is blocked instantly and cannot be used for transactions.
- Given a card is blocked, when the customer requests a replacement, then a new card is dispatched within 5-7 business days.

Module 9: Personal Loan Application

US-14 — Check loan eligibility

- Given a customer enters income and existing obligation details, when they request an eligibility check, then the system displays an indicative loan amount and interest rate.
- Given eligibility is displayed, when the customer proceeds, then they are directed to the full application form.

US-15 — Review loan applications

- Given an application is submitted, when it appears in the processing queue, then the officer can view all documents and eligibility data in one screen.
- Given the officer approves or rejects an application, when the decision is saved, then the customer is notified and the status updates in their application tracker.

US-16 — Track loan application status

- Given an application is submitted, when the customer views the tracker, then it shows the current stage (Submitted, Under Review, Approved, Disbursed, or Rejected).
- Given the status changes, when an update occurs, then the customer receives a notification.

Module 10: Notifications

US-17 — Transaction and reminder alerts

- Given a transaction occurs on the account, when it is processed, then a notification is sent within 1 minute via the customer's preferred channel.
- Given a customer has upcoming FD maturity or bill due dates, when the due date approaches, then a reminder is sent 3 days in advance.

Module 11: Customer Support

US-18 — Raise support ticket

- Given a customer submits a ticket with issue details, when it's created, then a ticket ID is generated and the customer can track its status.
- Given a ticket remains unresolved for 48 hours, when the SLA breach occurs, then it is auto-escalated to a senior support agent.

US-19 — Chatbot FAQ handling

- Given a customer asks a common question (e.g., "how to reset MFA"), when the chatbot identifies the intent, then it provides a resolution without human intervention.
- Given the chatbot cannot resolve a query, when it detects this, then it offers to escalate the issue to a live agent.

Module 12: Admin Portal

US-20 — Manage roles and permissions

- Given an administrator creates a new internal user, when a role is assigned, then access is restricted to permissions mapped to that role.

- Given a user's role changes, when updated, then access permissions are revised immediately.

US-21 — Platform health monitoring

- Given the admin portal loads, when the administrator views the monitoring dashboard, then it displays real-time uptime, error rate, and transaction volume metrics.
- Given a system anomaly is detected (e.g., error rate spike), when the threshold is breached, then an alert is sent to the IT team.

US-22 — Generate audit reports

- Given a compliance officer selects a date range, when they generate a report, then it includes onboarding, transaction, and MFA logs for that period.
- Given a report is generated, when exported, then it is available in PDF/Excel format.

US-23 — Fraud monitoring dashboard

- Given a transaction matches a fraud rule (e.g., unusual amount/location), when flagged, then it appears on the fraud dashboard with relevant details.
- Given an analyst reviews a flagged transaction, when they mark it as fraudulent, then the account is suspended pending further review.

US-24 — Loan queue with filters

- Given the officer opens the loan queue, when filters are applied (status, date, amount), then only matching applications are displayed.
- Given an application is selected, when opened, then all supporting documents and eligibility data are visible on one screen.

14. Process Flow

Process Name

Customer Onboarding — Branch-Based Account Opening (As-Is)

Objective

To document the current manual, branch-dependent process by which a retail customer opens a new savings account at ABC Bank Ltd. National Bank, from initial branch visit through account activation, debit card issuance, and internet banking access.

Actors

- **Customer** — individual seeking to open a new account
- **Customer Service Representative (CSR)** — first point of contact at the branch
- **Branch Officer** — verifies and approves the application
- **Operations Team** — performs backend data entry and card/internet banking requests
- **Core Banking System (CBS)** — legacy system of record for account creation

Step-by-Step Current Process (20 Steps)

1. Customer visits the branch during working hours with original ID and address proof.
2. Customer takes a token and waits in the general queue.
3. CSR greets the customer and understands the requirement (new account opening).
4. CSR manually verifies the customer's original identity and address proof documents (physical KYC check).
5. CSR hands over a blank paper account-opening application form to the customer.
6. Customer manually fills out the paper form (personal details, nominee, account type).
7. Customer attaches self-attested photocopies of ID proof, address proof, and a passport-size photograph.
8. CSR reviews the filled form for completeness and accuracy.
9. **Decision Point:** If the form is incomplete or documents are missing, CSR returns it to the customer for correction (loop back to Step 6).
10. If complete, CSR logs the application manually in a physical register and forwards it to the Branch Officer.
11. Branch Officer manually cross-checks the application against the submitted documents.
12. **Decision Point:** Branch Officer approves or rejects the application based on document authenticity and eligibility (e.g., age, existing account checks).
13. If rejected, the customer is informed verbally or via a follow-up call; process ends.
14. If approved, Branch Officer forwards the physical file to the Operations Team.

15. Operations Team manually keys the customer's data into the Core Banking System (CBS).
16. CBS generates a new account number and stores the customer record.
17. Operations Team submits a manual request to the card division for debit card issuance.
18. Debit card is printed centrally and dispatched to the customer's registered address via courier (typically 7–10 business days).
19. Operations Team separately submits a request to activate internet banking, which is provisioned manually within CBS.
20. Customer receives account details via a physical welcome letter and/or SMS notification once the account is active.

Decision Points

- Application form completeness check (Step 9)
- Branch Officer approval/rejection based on document verification (Step 12)

Manual Activities

- Physical KYC document verification
- Paper form filling by customer
- Manual logging in a physical register
- Manual cross-checking of documents by Branch Officer
- Manual data entry into CBS by Operations Team
- Manual, separate requests for debit card and internet banking activation

System Interactions

- CBS: customer record creation, account number generation
- Card printing/dispatch system: triggered manually, not integrated with CBS in real time
- Internet banking provisioning module: activated manually by Operations Team, not automatic on account creation
- SMS/notification system: triggered only after account is fully active

Pain Points and Bottlenecks

- Customer must visit the branch physically and cannot begin the process online.
- Long, unpredictable wait times due to token-based queueing.
- Form errors discovered only after manual review, requiring resubmission and repeat visits.
- No real-time verification of ID/address proof — relies entirely on human judgment.
- Multiple handoffs (CSR → Branch Officer → Operations Team) with no shared digital workflow, increasing turnaround time and risk of lost files.
- Debit card and internet banking activation are handled as separate, sequential manual requests instead of running in parallel with account creation.
- No visibility for the customer into application status between submission and account activation.
- Physical document storage creates compliance and retrieval risk.

Business Rules

- Customer must be physically present with original documents for KYC — no remote or digital verification permitted.
- Application form must be complete before being forwarded to the Branch Officer.
- Only the Branch Officer can approve or reject a new account application.
- Debit card and internet banking requests cannot be initiated until the account is confirmed created in CBS.
- Customer must be 18 years or older to open an individual account.

Inputs and Outputs

Inputs: Original ID proof, original address proof, passport-size photograph, filled paper application form, nominee details

Outputs: Active bank account with account number, physical welcome letter, debit card (dispatched separately), internet banking credentials, SMS/notification confirming activation

Opportunities for Improvement

- Replace physical KYC with digital eKYC (Aadhaar/PAN-based verification with video KYC).

- Eliminate paper forms in favor of a digital application captured directly into the system.
- Enable parallel processing of debit card and internet banking requests instead of sequential handling.
- Provide real-time application status visibility to the customer.
- Reduce manual handoffs between CSR, Branch Officer, and Operations Team through a shared digital workflow queue.

BPMN-Style Process Flow Description

Pool: ABC Bank Ltd. National Bank — Branch Account Opening (As-Is)

Lane: Customer

- Start Event: Customer arrives at branch
- Task: Take token, wait in queue
- Task: Provide original documents for KYC
- Task: Fill paper application form
- Task: Attach photocopies and photograph

Lane: Customer Service Representative (CSR)

- Task: Perform manual KYC verification
- Task: Issue application form
- Task: Review form completeness
- Gateway (Exclusive): Form complete?
 - No → Task: Return form to customer (loop to Customer lane)
 - Yes → Task: Log application, forward to Branch Officer

Lane: Branch Officer

- Task: Cross-check documents against application
- Gateway (Exclusive): Application approved?
 - No → Task: Notify customer of rejection → End Event: Application rejected
 - Yes → Task: Forward file to Operations Team

Lane: Operations Team

- Task: Enter customer data into CBS

- Task: Request debit card issuance
- Task: Request internet banking activation

Lane: Core Banking System (CBS)

- Task: Generate account number and create record
- Task: Provision internet banking access
- End Event: Notification sent to customer, account active

Pain Points Identified by the Business Analyst

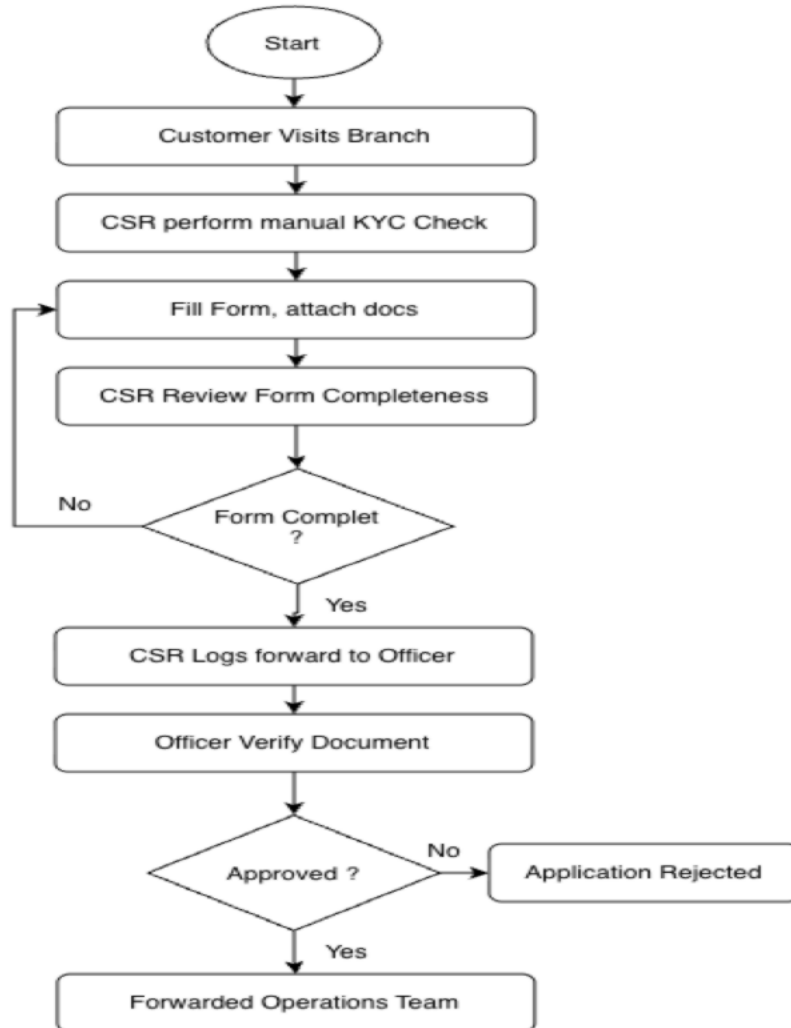
1. Physical presence requirement excludes customers who cannot easily visit a branch, contributing to the onboarding delays flagged in the Business Problem.
2. Sequential, non-integrated handoffs between CSR, Branch Officer, and Operations Team introduce delay and risk of file loss with no digital audit trail.
3. Manual KYC and data entry are error-prone and slow compared to automated verification against government databases.
4. Debit card and internet banking are treated as afterthoughts rather than part of a single onboarding transaction, extending time-to-first-use well beyond account creation.
5. Customers have no self-service visibility into where their application stands, driving avoidable branch and call-centre follow-ups.

Recommendations for the To-Be Process

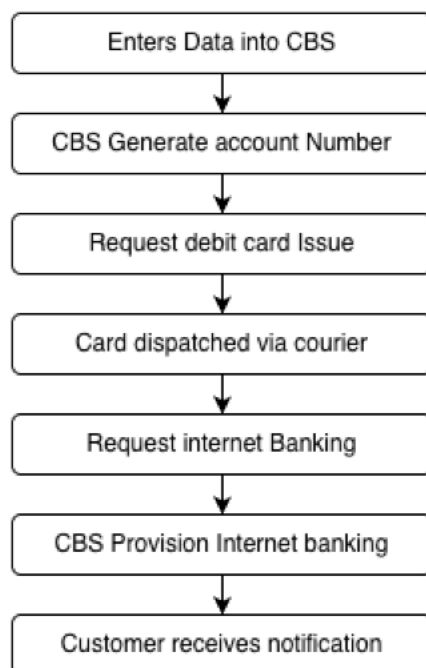
- Replace manual KYC with digital eKYC (Aadhaar/PAN validation + video KYC) to remove the branch dependency entirely for standard cases.
- Digitize the application form so customer-entered data flows directly into the platform, removing manual re-keying by Operations Team.
- Introduce a risk-based approval model: auto-approve standard applications, route only flagged/mismatched cases to manual compliance review.
- Trigger debit card issuance and internet banking activation automatically and in parallel upon account creation, rather than as separate sequential requests.
- Provide customers a real-time application status tracker to eliminate status-check calls and visits.

As-Is Process Flow — Part 1: Branch Application & Approval

Part 1: Branch Application & Approval



As-Is Process Flow — Part 2: Backend Processing & Activation



Process Name

Digital Customer Onboarding — Self-Service Account Opening (To-Be)

Business Objective

To enable a retail customer to open a fully functional bank account — including internet banking and a debit card — entirely through the web or mobile app, without visiting a branch, using automated eKYC verification and system-to-system integration to reduce onboarding time from 2-3 days to same-day activation.

Actors

- **Customer** — individual opening a new account via web or mobile app
- **Digital Banking Platform** — the customer-facing application orchestrating the journey
- **eKYC Service** — third-party/government-integrated identity verification service (Aadhaar/PAN, video KYC)
- **Core Banking System (CBS)** — system of record for account creation
- **Notification Service** — sends SMS/email/in-app alerts

- **Bank Administrator** — reviews only flagged/exception cases

Step-by-Step Future Process (18 Steps)

1. Customer downloads the app or visits the web platform and selects "Open New Account."
2. Customer enters mobile number and email address.
3. Platform sends OTP to mobile number and email simultaneously.
4. Customer enters both OTPs to verify contact details.
5. **Decision Point:** Are both OTPs valid? If invalid, customer is prompted to re-enter or request a resend (loop to Step 3).
6. Customer enters PAN and Aadhaar number.
7. Platform calls the eKYC Service to validate identity against UIDAI/NSDL in real time.
8. Customer uploads a photo ID and completes a live video KYC session within the app.
9. eKYC Service cross-checks the video KYC capture against the submitted ID document (face match, liveness check).
10. **Decision Point:** Does eKYC pass validation? If a mismatch or low-confidence match is detected, the application is routed to the Bank Administrator for manual review; customer is notified their application is under review (exception path).
11. If eKYC passes automatically, the platform proceeds to digital account opening — customer selects account type and reviews terms.
12. Customer digitally accepts terms and conditions and submits the application.
13. Platform sends the validated application data to the Core Banking System via API.
14. CBS automatically creates the account and generates an account number in real time.
15. Platform triggers instant Internet Banking activation upon account creation confirmation from CBS.
16. Customer sets up login credentials and registers for Multi-Factor Authentication (OTP and/or biometric).
17. Platform submits a parallel automated request for debit card issuance (physical + optional instant virtual card).
18. Notification Service sends a real-time welcome email and SMS confirming the account number, internet banking activation, and card dispatch status.

Decision Points

- OTP validity check for mobile and email (Step 5)

- eKYC pass/fail validation, determining automated approval vs. manual review routing (Step 10)

Automated System Activities

- OTP generation and validation
- Real-time PAN/Aadhaar validation against government databases
- Face match and liveness detection during video KYC
- Automatic account number generation in CBS
- Automatic Internet Banking provisioning
- Automatic debit card issuance request
- Automated real-time notifications at each key milestone

System Integrations

- Digital Banking Platform ↔ eKYC Service (identity verification API)
- Digital Banking Platform ↔ Core Banking System (account creation API)
- Digital Banking Platform ↔ Notification Service (SMS/email/push gateway)
- Core Banking System ↔ Card Management System (debit card issuance trigger)
- Digital Banking Platform ↔ MFA Provider (OTP/biometric authentication service)

Business Rules

- Mobile number and email must each be verified via OTP before proceeding to eKYC.
- Customer must be 18 years or older; the system shall auto-reject underage applicants at the PAN/Aadhaar validation step.
- Video KYC must achieve a minimum face-match confidence threshold; below this, the application routes to manual review, never auto-rejects.
- Internet Banking shall only activate after CBS confirms successful account creation.
- MFA registration is mandatory before first login is permitted.
- Debit card issuance request shall trigger automatically and in parallel with Internet Banking activation, not sequentially.

Inputs and Outputs

Inputs: Mobile number, email address, PAN, Aadhaar number, live video KYC capture, digitally accepted terms and conditions

Outputs: Active bank account with account number, activated Internet Banking credentials, MFA registration, debit card dispatch request confirmation, real-time welcome email/SMS

Exception Scenarios

- **OTP not received:** Customer can request a resend after a cooldown period; repeated failures direct the customer to alternate verification (registered branch or support).
- **eKYC face-match failure:** Application routed to Bank Administrator queue for manual review; customer notified of delay, not rejection.
- **PAN/Aadhaar validation service downtime:** Application is held in a pending state with customer notification; automatic retry attempted before manual fallback.
- **CBS account creation failure:** Platform retries automatically; if unresolved, the case is escalated to the Bank Administrator with the customer notified of the delay.
- **Underage applicant:** System auto-rejects at PAN/Aadhaar validation with a clear on-screen message; no manual review needed

BPMN-Style Process Flow Description

Pool: ABC Bank Ltd. National Bank — Digital Customer Onboarding (To-Be)

Lane: Customer

- Start Event: Customer opens app/web platform
- Task: Enter mobile number and email
- Task: Enter OTPs received
- Task: Enter PAN and Aadhaar
- Task: Complete video KYC
- Task: Select account type, accept terms
- Task: Set up MFA credentials

Lane: Digital Banking Platform

- Task: Send OTP to mobile and email

- Gateway (Exclusive): OTPs valid?
 - No → Task: Prompt resend (loop to Customer lane)
 - Yes → Task: Call eKYC Service
- Task: Submit application data to CBS
- Task: Trigger Internet Banking activation
- Task: Trigger parallel debit card request

Lane: eKYC Service

- Task: Validate PAN/Aadhaar against government database
- Task: Perform face-match and liveness check on video KYC
- Gateway (Exclusive): eKYC passed?
 - No → Task: Route to Bank Administrator (manual review)
 - Yes → Task: Return validation success to platform

Lane: Core Banking System

- Task: Create account record
- Task: Generate account number
- Task: Confirm creation to platform

Lane: Notification Service

- Task: Send OTP messages
- Task: Send welcome email/SMS with account and activation details
- End Event: Customer onboarded and notified

Lane: Bank Administrator

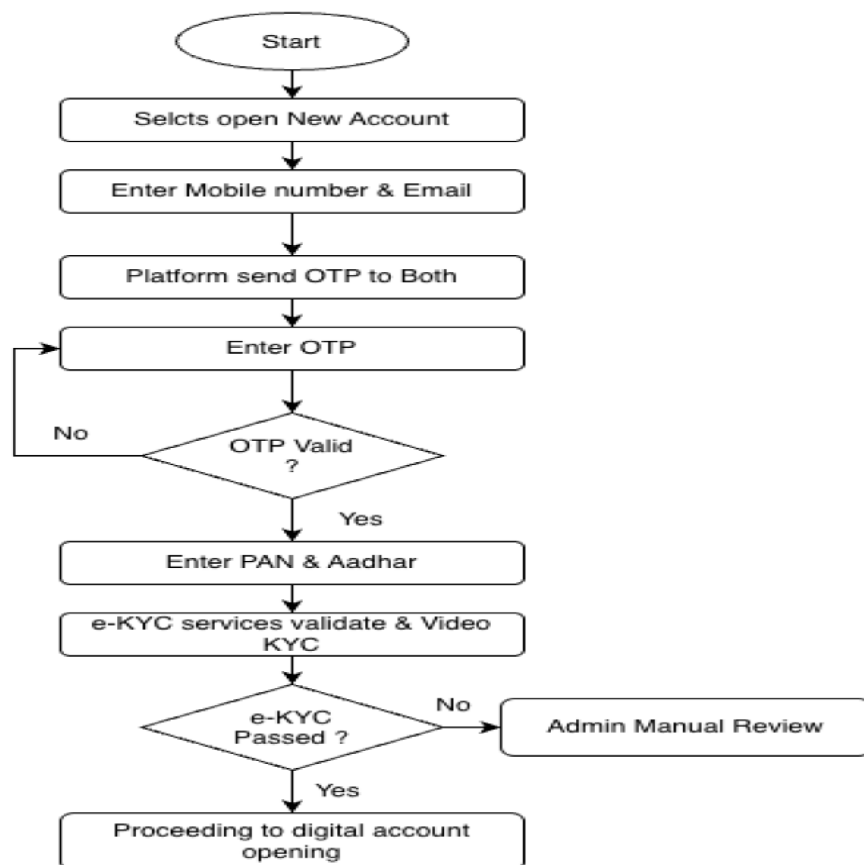
- Task: Review flagged eKYC exceptions
- Gateway (Exclusive): Manual approval granted?
 - No → End Event: Application rejected, customer notified
 - Yes → Task: Resume automated flow (rejoin main path)

Comparison: As-Is vs. To-Be

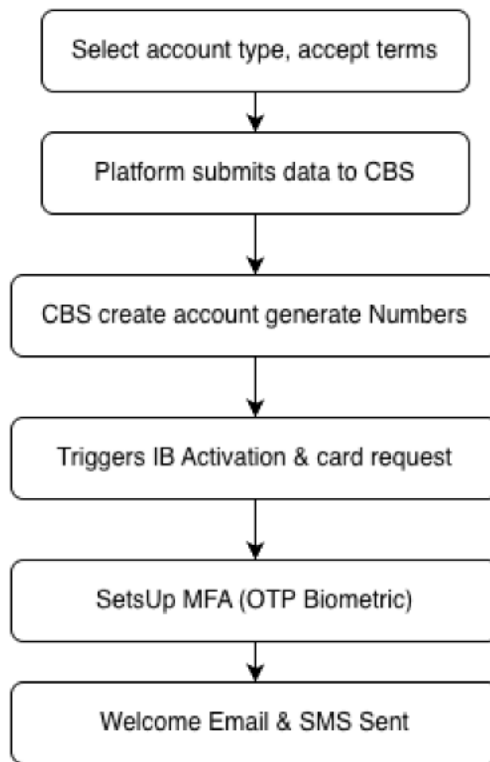
Aspect	As-Is	To-Be
Channel	Branch only	Web and mobile app

Identity verification	Manual, physical document check	Automated eKYC with video verification
Application format	Paper form	Fully digital
Turnaround time	2-3 working days	Same-day, often minutes
Human involvement	CSR, Branch Officer, Operations Team at every step	System-automated; human involved only for flagged exceptions
Card & Internet Banking activation	Sequential, separately requested	Parallel, automatic
Authentication	Single-factor login	Mandatory MFA
Status visibility	None until completion	Real-time in-app and notification updates.
Compliance/audit trail	Manual register, physical files	Fully digital, system-logged

To-Be Process Flow — Part 1: Verification & eKYC



To-Be Process Flow — Part 2: Account Creation & Activation



15. UAT Scenarios

Module 1: Customer Registration & eKYC

Fields	Details
Scenario ID	UAT-01
Scenario Name	Register New Account via eKYC
Preconditions	Customer has valid Aadhaar/PAN and no existing account
Steps	1. Open registration page 2. Enter Aadhaar/PAN details 3. Upload identity documents 4. Complete video KYC 5. Submit application
Expected Results	Account created, status = "Active" within same business day
Actual Result	Account created and activated same day as expected
Status	Pass

Scenario ID	UAT-02
Scenario Name	Reject Mismatched KYC Document

Preconditions	Customer on registration form with document upload step
Steps	1. Upload document with name/DOB mismatch 2. Submit application 3. Check application status
Expected Results	Application routed to manual review queue, customer notified
Actual Result	Application flagged and routed for compliance review as expected
Status	Pass

Module 2: Login with MFA

Scenario ID	UAT-03
Scenario Name	Login with OTP Verification
Preconditions	Customer has an active account with registered mobile number
Steps	1. Enter username and password 2. Select OTP as second factor 3. Enter received OTP 4. Submit
Expected Results	Login successful, customer redirected to dashboard
Actual Result	Login completed successfully after OTP verification
Status	Pass

Scenario ID	UAT-04
Scenario Name	Account Lockout After Failed OTP Attempts
Preconditions	Customer at OTP entry step during login
Steps	1. Enter incorrect OTP 2. Repeat 3 times 3. Attempt further login
Expected Results	Account locked after 3rd failed attempt, customer notified
Actual Result	Account locked as expected; lockout notification sent
Status	Pass

Module 3: Dashboard

Fields	Details
Scenario ID	UAT-05.
Scenario Name	View Consolidated Dashboard on Login
Preconditions	Customer has active accounts and at least one linked card
Steps	1. Login to platform 2. Land on dashboard
Expected Results	Dashboard shows balances, linked cards, and 5 most recent transactions
Actual Result	Dashboard displayed all elements correctly

Status	Pass
--------	------

Scenario ID	UAT-06
Scenario Name	Display Active Alerts on Dashboard
Preconditions	Customer has a low balance or upcoming FD maturity
Steps	1. Login to platform 2. View dashboard top section
Expected Results	Relevant alert displayed prominently at top of dashboard
Actual Result	Low balance alert displayed correctly at top
Status	Pass

Module 4: Account Summary

Scenario ID	UAT-07
Scenario Name	Download Account Statement
Preconditions	Customer has transaction history for the selected period
Steps	1. Navigate to Account Summary 2. Select date range 3. Click "Generate Statement" 4. Download PDF
Expected Results	Statement generated within 10 seconds and downloadable as PDF
Actual Result	Statement generated in 6 seconds and downloaded successfully
Status	Pass

Fields	Details
Scenario ID	UAT-08
Scenario Name	Search Transaction History by Type
Preconditions	Customer account has mixed transaction types (debit/credit)
Steps	1. Navigate to Account Summary 2. Apply filter by transaction type 3. View results
Expected Results	Only transactions matching selected type are displayed
Actual Result	Filtered results matched selected transaction type
Status	Pass

Module 5: Fund Transfer

Scenario ID	UAT-09
Scenario Name	Transfer Funds via IMPS to Registered Beneficiary
Preconditions	Beneficiary added and verified, sufficient balance available

Steps	1. Navigate to Fund Transfer 2. Select beneficiary 3. Enter amount 4. Confirm transfer
Expected Results	Transfer processed successfully, confirmation displayed
Actual Result	Transfer completed and confirmation shown as expected
Status	Pass

Scenario ID	UAT-10
Scenario Name	Reject Transfer Exceeding Daily Limit
Preconditions	Customer's daily transfer limit already near maximum
Steps	1. Navigate to Fund Transfer 2. Enter amount exceeding daily limit 3. Submit
Expected Results	System rejects transfer with explanatory error message
Actual Result	Transfer rejected with correct limit-exceeded message
Status	Pass

Module 6: Bill Payment

Scenario ID	UAT-11
Scenario Name	Register Biller and Pay Bill
Preconditions	Customer has valid biller account number
Steps	1. Navigate to Bill Payment 2. Search and register biller 3. Enter amount 4. Confirm payment
Expected Results	Payment completed, receipt generated and stored in history
Actual Result	Payment successful, receipt generated as expected
Status	Pass

Fields	Details
Scenario ID	UAT-12
Scenario Name	View Bill Payment History
Preconditions	Customer has made at least one prior bill payment
Steps	1. Navigate to Bill Payment 2. Open "Payment History" tab
Expected Results	All past payments listed with date, biller, and amount
Actual Result	Payment history displayed correctly
Status	Pass

Module 7: Fixed Deposit

Scenario ID	UAT-13
Scenario Name	Book New Fixed Deposit
Preconditions	Customer has sufficient balance in linked account
Steps	1. Navigate to Fixed Deposit 2. Enter amount and tenure 3. Review interest rate and maturity value 4. Confirm booking
Expected Results	FD booked, funds debited, e-certificate generated
Actual Result	FD booked successfully with certificate issued
Status	Pass

Scenario ID	UAT-14
Scenario Name	Request Premature FD Closure
Preconditions	Customer has an active fixed deposit
Steps	1. Navigate to active FD 2. Select "Close Prematurely" 3. Review penalty and payout 4. Confirm closure
Expected Results	Closure processed, funds credited within 1 business day
Actual Result	Closure processed and funds credited next business day
Status	Pass

Module 8: Credit Card Management

Fields	Details
Scenario ID	UAT-15
Scenario Name	Activate New Credit Card
Preconditions	Customer has received a new physical card
Steps	1. Navigate to Credit Card Management 2. Enter card and CVV details 3. Complete MFA verification
Expected Results	Card activated, limit and available credit displayed
Actual Result	Card activated and details displayed correctly
Status	Pass

Scenario ID	UAT-16
Scenario Name	Block Lost Card and Request Replacement
Preconditions	Customer has an active credit card
Steps	1. Navigate to Credit Card Management 2. Select "Block Card"

	3. Confirm via MFA 4. Request replacement
Expected Results	Card blocked instantly, replacement dispatched within 5-7 days
Actual Result	Card blocked immediately, replacement request logged
Status	Pass

Module 9: Personal Loan Application

Scenario ID	UAT-17
Scenario Name	Check Loan Eligibility
Preconditions	Customer has active account with income details available
Steps	1. Navigate to Personal Loan 2. Enter income and obligations 3. Submit for eligibility check
Expected Results	Indicative loan amount and interest rate displayed
Actual Result	Eligibility results displayed as expected
Status	Pass

Scenario ID	UAT-18
Scenario Name	Track Loan Application Status
Preconditions	Customer has submitted a loan application
Steps	1. Navigate to "My Applications" 2. Select active application 3. View status tracker
Expected Results	Current stage displayed (Submitted/Under Review/Approved/Disbursed/Rejected)
Actual Result	Status tracker showed correct current stage
Status	Pass

Module 10: Notifications

Fields	Details
Scenario ID	UAT-19
Scenario Name	Receive Transaction Alert
Preconditions	Customer has notifications enabled
Steps	1. Perform a fund transfer 2. Check SMS/email/in-app notification
Expected Results	Notification received within 1 minute of transaction
Actual Result	Notification received within 30 seconds
Status	Pass

Scenario ID	UAT-20
Scenario Name	Receive FD Maturity Reminder
Preconditions	Customer has an FD maturing within 3 days
Steps	1. Wait for reminder trigger date 2. Check notification channel
Expected Results	Reminder sent 3 days before maturity date
Actual Result	Reminder received on expected date
Status	Pass

Module 11: Customer Support

Scenario ID	UAT-21
Scenario Name	Raise Support Ticket
Preconditions	Customer is logged in and facing a service issue
Steps	1. Navigate to Customer Support 2. Select "Raise Ticket" 3. Enter issue details 4. Submit
Expected Results	Ticket ID generated, status trackable by customer
Actual Result	Ticket created with valid ID and trackable status
Status	Pass

Fields	Details
Scenario ID	UAT-22
Scenario Name	Resolve Query via Chatbot
Preconditions	Customer has a common query (e.g., MFA reset)
Steps	1. Open chatbot 2. Type query 3. Review chatbot response
Expected Results	Chatbot resolves query without human intervention
Actual Result	Chatbot provided correct resolution steps
Status	Pass

Module 12: Admin Portal

Scenario ID	UAT-23
Scenario Name	Assign Role-Based Access to Internal User
Preconditions	Administrator is logged into Admin Portal
Steps	1. Navigate to User Management 2. Create new internal user 3. Assign role

	4. Save
Expected Results	User access restricted to permissions mapped to assigned role
Actual Result	Role-based access applied correctly
Status	Pass

Scenario ID	UAT-24
Scenario Name	Generate Compliance Audit Report
Preconditions	Compliance officer logged into Admin Portal
Steps	1. Navigate to Reports 2. Select date range 3. Generate report 4. Export as PDF
Expected Results	Report includes onboarding, transaction, and MFA logs for selected period
Actual Result	Report generated and exported correctly
Status	Pass

16. Challenge & Solutions

Challenge 1: Balancing security friction with usability in MFA

Introducing mandatory MFA on every login risked frustrating customers and increasing drop-off, while the Fraud & Risk team pushed for the strictest possible controls.

Solution: Ran a joint workshop with Fraud & Risk and IT to agree on a one-extra-step limit (NFR-07), offering both OTP and biometric as interchangeable second factors so customers could choose the faster option without weakening security.

Challenge 2: Conflicting requirements between Compliance and Customer

Experience

Compliance wanted extensive manual document review for every eKYC submission, which would have defeated the goal of same-day onboarding.

Solution: Proposed a risk-based validation model — automated checks for standard cases, with manual review triggered only when the system detects a name/DOB mismatch or other red flag (FR-02). This preserved compliance oversight without slowing down the majority of applications.

Challenge 3: Legacy core banking system (CBS) integration constraints

Early technical workshops revealed the existing CBS did not expose real-time APIs for all required data (e.g., FD interest calculations), risking delays to the Account Summary and

Fixed Deposit modules.

Solution: Worked with the IT team to sequence delivery — modules dependent on available APIs were prioritized first, while a phased API development plan was agreed with the CBS vendor for the remaining data points, and this dependency was documented in the Assumptions & Constraints section.

Challenge 4: Scope creep from stakeholder requests

Multiple stakeholder groups (Relationship Managers, Loan Processing Team) requested additional features during workshops that fell outside the board-approved roadmap, such as investment product cross-selling.

Solution: Introduced a formal change request process tied to the Scope document, requiring BA sign-off before any new feature was added — keeping the 12-module scope intact while logging out-of-scope requests for a potential future phase.

Challenge 5: Fragmented input from a wide stakeholder base

With 10 distinct stakeholder groups involved (per the Stakeholder Analysis), individual interviews and workshops surfaced overlapping or contradictory requirements — particularly between Customer Support and Fraud & Risk on how much account information a support executive should be able to view.

Solution: Consolidated conflicting inputs into a single requirements review workshop where both teams jointly defined role-based access boundaries, later reflected in the Admin Portal's role-based access control requirement (FR-16).

Challenge 6: Validating requirements without a working prototype

Business stakeholders struggled to visualize how digital onboarding or loan tracking would actually function from written requirements alone.

Solution: Used low-fidelity wireframe walkthroughs during the elicitation phase (per the Requirement Gathering Approach) to validate assumptions before finalizing the FR/NFR document, reducing rework during later UAT.

Challenge 7: Migrating legacy customer data without service disruption

Existing customer records needed to move from three legacy systems into the unified platform, with risk of data loss or mismatched account histories.

Solution: Recommended a parallel-run period with data reconciliation checks before final cutover, documented as a mitigation under the Risks register, ensuring customers saw no discrepancy in balances or transaction history at go-live.

17. BA tools & Techniques

Throughout this project, a mix of established Business Analysis techniques was applied at each stage of the requirement lifecycle — from initial elicitation through documentation, validation, and testing — rather than relying on a single method for the entire engagement.

The requirement gathering phase started with **document analysis**, reviewing the bank's existing net banking specifications, RBI KYC guidelines, and historical customer complaint logs to understand where the current process was falling short. This was followed by **stakeholder interviews** with the Loan Processing, Compliance, and IT teams, each conducted one-on-one to capture role-specific pain points and constraints that wouldn't surface in a group setting. Where cross-functional alignment was needed — particularly between the Fraud & Risk and IT teams on MFA design — **facilitated workshops** were used instead, since those decisions required real-time negotiation rather than sequential input.

To understand the customer experience directly, **process observation** was used by shadowing customer support executives during live calls, which surfaced which queries were genuinely complex versus which ones were simply repetitive and suitable for self-service. This was supplemented with a customer **survey** to quantify digital adoption barriers across a broader sample than interviews alone could reach.

Once early requirements took shape, **low-fidelity prototype walkthroughs** were used to validate assumptions with stakeholders before anything was finalized in writing — this caught several usability concerns (like MFA friction) early enough to resolve without rework.

For documentation and analysis, the project relied on a standard set of techniques: **MoSCoW-style prioritization** to rank requirements as High or Medium priority, **user story mapping** to break each of the 12 modules into role-based stories, and the **Given-When-Then (Gherkin) format** to write acceptance criteria that could be tested unambiguously. **As-Is/To-Be process mapping** was used to visually contrast the current branch-dependent process against the proposed digital journey, which proved useful both for stakeholder buy-in and for scoping the technical work. A **requirement traceability matrix** tied every user story back to its originating requirement, business objective, and stakeholder — so nothing in the final deliverable could be traced to "nowhere."

Risk was handled through a straightforward **risk register** — documenting impact, likelihood, and mitigation for each identified risk — and **root cause analysis** was applied specifically to the Business Problem, to make sure the platform's scope was addressing actual causes (fragmented systems, manual handoffs) rather than surface symptoms.

Finally, before sign-off, **UAT scenario design** translated the acceptance criteria into testable scenarios, and a **business impact analysis** was used to frame the completed requirements in terms of customer, operational, financial, and risk outcomes — the language leadership actually needed to approve the investment.

Taken together, these techniques reflect a fairly standard BA toolkit, but the sequencing mattered: elicitation techniques came first to surface real problems, modeling and documentation techniques structured that input into buildable requirements, and validation techniques (prototyping, traceability, UAT) made sure nothing got lost between those two stages.